

EX NO:2

Chebyshev IIR Filter

2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter

Objectives:

1. To design and analyze the magnitude response, phase response, impulse response, and group delay characteristics of the Chebyshev Type-I IIR High-pass filter for the given passband and stopband specifications.
2. To analyze the impulse response, and group delay characteristics of the Chebyshev Type-I IIR high-pass filter for the given passband and stopband specifications.
3. To examine the pole-zero plot and verify the stability and performance of the designed high-pass filter.
4. To determine the minimum filter order and cutoff frequency required to satisfy the design constraints.

Procedure:

1. Open MATLAB and specify the passband frequency, stopband frequency, passband ripple, stopband attenuation, and sampling frequency.
2. Normalize the frequencies with respect to the Nyquist frequency.
3. Determine the minimum filter order and cutoff frequency using `cheb1ord`.
4. Design the Chebyshev Type-I IIR High-Pass Filter using `cheby1`.
5. Obtain the filter coefficients and frequency response using `freqz`.
6. Plot and analyze the magnitude and phase responses.
7. Obtain and analyze the impulse response using `impz`.
8. Determine the group delay response using `grpdelay`.
9. Plot the pole-zero diagram using `zplane`.
10. Verify stability by ensuring that all poles lie inside the unit circle and record the observations.

Program (MATLAB Code):

```
clc;  
  
clear;
```

close all;

% Specifications

Rp = 1; % Passband Ripple (dB)

Rs = 40; % Stopband Attenuation (dB)

Wp = 0.5; % Passband Edge

EX NO:2 Chebyshev IIR Filter

DATE: 2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass

Filter

Ws = 0.35; % Stopband Edge

% Find Filter Order

[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

% Design High-Pass Filter

[b,a] = cheby1(n,Rp,Wn,'high');

% Frequency Response

figure;

freqz(b,a);

title('Frequency Response');

% Impulse Response

figure;

impz(b,a);

title('Impulse Response');

% Group Delay

figure;

```
grpdelay(b,a);
```

```
title('Group Delay');
```

```
% Pole-Zero Plot
```

```
figure;
```

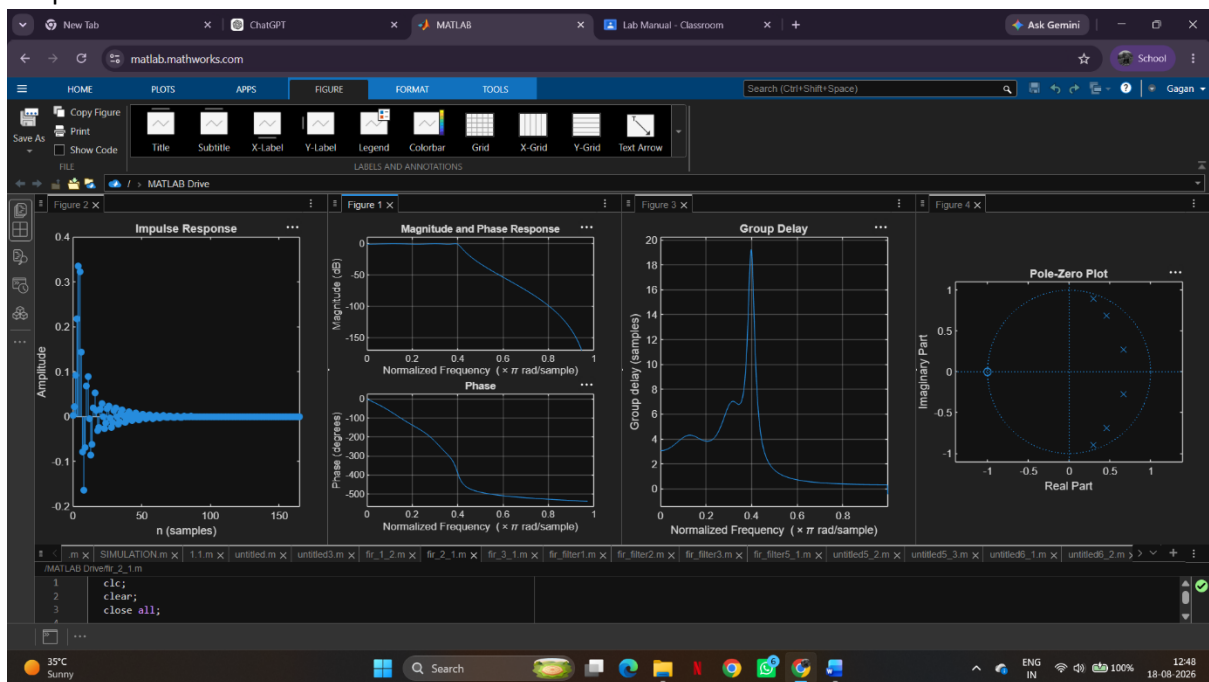
```
zplane(b,a);
```

```
title('Pole-Zero Plot');
```

```
fprintf('Filter Order = %d\n',n);
```

```
fprintf('Cutoff Frequency = %.4f\n',Wn);
```

Output:



Filter Order = 6

Cutoff Frequency = 0.5000

Result:

1. The Chebyshev Type-I IIR High-pass filter was successfully designed, and its magnitude response, phase response, impulse response, and group delay were analyzed.
2. The impulse response and group delay characteristics were obtained and found to satisfy the given specifications.

3. The pole-zero plot confirmed that all poles lie inside the unit circle, verifying the stability and proper performance of the filter.
4. The minimum filter order and cutoff frequency were successfully determined and found to satisfy the given design specifications.